Error-Corrected Heterogeneous Treatment Effects: When LLM Annotation Debiasing Hurts

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Abstract

We prove that correcting LLM annotation error with a single pooled confusion matrix—standard practice when LLM labels serve as effect modifiers in heterogeneous treatment effect (HTE) analysis—introduces subgroup-level bias exactly proportional to the product of misclassification heterogeneity and CATE heterogeneity ($\Delta\pi \times \Delta\tau$), an exact population-level identity verified at $R^2 = 0.994$ (K=2). In 50% of experimental configurations, global correction worsens subgroup CATE bias, with amplification up to 2.86× (K=2, difference-in-means) and 5.9× (K=4, CausalForest). A natural solution enabled by this insight, Error-Corrected Heterogeneous Treatment Effects (EC-HTE), combines hierarchical Bayesian subgroup-stratified confusion matrix estimation with CausalForestDML, reducing |bias| from 0.059 to 0.010 while maintaining nominal coverage (≥0.91 across all configurations) under extreme heterogeneity (K=2); a computable diagnostic (AUC = 0.86) identifies when global correction increases bias from per-subgroup misclassification rates alone, actionable without adopting EC-HTE. Composed with Prediction-Powered Conditional Inference (PPCI) for dual covariate/outcome error, the two corrections are each necessary and jointly sufficient, reducing |bias| from 0.179 to 0.007. The correction is estimator-agnostic and robust to non-additive treatment effects: replacing CausalForest with the R-learner yields comparable bias reduction (89%), and a multiplicative DGP yields nominal coverage (0.955). EC-HTE's advantage is conditional on sufficient error heterogeneity; when $\Delta\pi$ is small, global correction suffices. We validate across 520 synthetic configurations (100 Monte Carlo runs each), a semi-synthetic TweetEval experiment (N=5,000), and seven LLMs (four open-source 7B–9B, three proprietary) across two NLP datasets confirming $\Delta\pi$ heterogeneity on real annotations.

1 Introduction

Large language models (LLMs) now routinely replace human annotators in computational social science (CSS), producing labels for sentiment, hate speech, stance, and political discourse at costs orders of magnitude below manual coding (Gilardi et al., 2023; Törnberg, 2024; Ziems et al., 2024). Yet these annotations are not uniformly accurate. Ashwin et al. (2025) document that LLM misclassification rates vary by up to 5× across demographic subgroups defined by refugee status, gender, and education level. In a pilot study, GPT-4.1 achieves 85.8% binary accuracy on TweetEval sentiment, but error rates range from 10.2% to 20.0% across text-feature subgroups ($\Delta\pi = 0.098$), within the range assumed in our synthetic DGPs ($\Delta\pi = 0.05$–0.20). When such annotations serve as effect modifiers in heterogeneous treatment effect (HTE) analysis, subgroup-specific errors propagate directly into conditional average treatment effect (CATE) estimates. The resulting bias is not random noise that averages out; it is structured by the same demographic axes that define the research question. Standard correction procedures, which estimate a single confusion matrix and apply its inverse uniformly, cannot account for this heterogeneity.

We formalize this problem and discover a previously unrecognized bias mechanism. When a pooled confusion matrix is applied to correct subgroup CATEs, it misattributes error across subgroups with an exact product form: the bias in subgroup s equals $\delta_s \cdot \Delta\tau/(1 - 2\bar{\pi})$, where δ_s is the deviation from the pooled error rate and $\Delta\tau$ is the CATE heterogeneity (Theorem 1). This is not a bound but an exact population-level identity, verified empirically with $R^2 = 0.994$ (K=2) and $R^2 = 0.974$ (K=4). A direct consequence is that global correction can be worse than no correction at all: in 50% of our experimental configurations, global correction increases absolute bias relative

to the naive estimator, with amplification reaching 2.86× (K=2) and 5.9× (K=4 with CausalForest). Error-Corrected Heterogeneous Treatment Effects (EC-HTE) addresses this through hierarchical Bayesian estimation of subgroup-stratified confusion matrices combined with CausalForestDML, reducing |bias| from 0.059 to 0.010 under extreme heterogeneity (K=2). The bias decomposition identity also yields a $\Delta\tau$-free diagnostic (Proposition 3): a condition on per-subgroup misclassification rates alone predicts when global correction will increase bias (AUC = 0.86)—independently useful even without adopting EC-HTE, since practitioners can flag harmful correction from a pilot sample before estimating any treatment effects. EC-HTE's advantage is conditional on sufficient error heterogeneity and expert budget; when $\Delta\pi$ is small, global correction suffices and EC-HTE's variance cost is not justified (Figure 1).

This problem matters because subgroup CATEs inform resource allocation decisions: which communities benefit most from policy interventions, where programs should be targeted, and how reforms affect different groups. Biased CATE estimates can reverse the sign of apparent treatment heterogeneity, leading practitioners to allocate resources away from communities that benefit most. The failure mode is subtle: a researcher who corrects for annotation noise using a pooled confusion matrix may introduce more bias than doing nothing, precisely in the low-error subgroups where original labels were most trustworthy. As LLM annotation pipelines replace manual coding across the social sciences, heterogeneous classification errors will propagate to an increasing share of causal analyses.

Several methods address measurement error in downstream statistical analyses, but none handles the specific failure mode we identify. Prediction-Powered Inference (PPI; Angelopoulos et al. 2023) and Design-based Supervised Learning (Egami et al., 2024) correct population-level aggregates by combining predicted and gold-standard labels, assuming homogeneous prediction error across subgroups. These methods produce valid confidence intervals for marginal quantities but do not extend to subgroup-conditional estimands. Prediction-Powered Conditional Inference (PPCI; Sui and Candès 2026) extends PPI to conditional means but targets outcome-side prediction error rather than covariate-side misclassification; it corrects $\hat{Y}$ given true Z, not the reverse. Mozer (2026) prove that arm-specific rectifiers fix average treatment effect

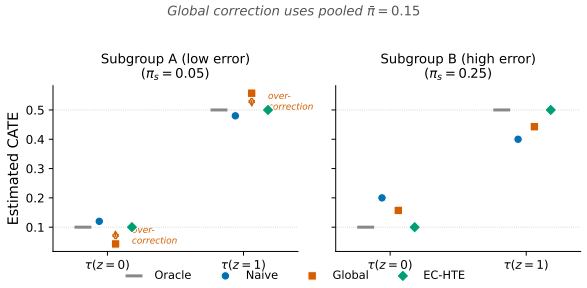


Figure 1: Bias mechanism under global vs. subgroup-stratified correction. Global correction applies the pooled error rate $\bar{\pi}$ uniformly, over-correcting low-error subgroups (introducing bias where none existed) and under-correcting high-error subgroups. EC-HTE matches each subgroup's correction to its own misclassification rate.

bias under differential prediction error, but their analysis is limited to population averages and does not address heterogeneity estimation. ForestIV (Bu et al., 2023) handles ML-covariate measurement error in regression forests but targets the outcome model rather than the treatment effect heterogeneity structure. The gap is specific: no existing method corrects bias when the mismeasured variable is the one that defines the heterogeneity structure itself, i.e., the effect modifier whose levels index the CATE function.

Our contributions are:

- A bias decomposition identity (Theorem 1) proving that global correction introduces bias proportional to $\Delta\pi \times \Delta\tau$, with a $\Delta\tau$-free diagnostic (Proposition 3; AUC = 0.86) that flags when correction is harmful using only per-subgroup misclassification rates—independently actionable without EC-HTE (Sections 3, 5.4).
- EC-HTE: hierarchical Bayesian subgroup-stratified confusion matrix estimation, compatible with any CATE estimator (CausalForestDML, R-learner) and composable with PPCI for dual covariate/outcome error (|bias|: 0.179 → 0.007), validated across 520 synthetic configurations, a semi-synthetic TweetEval experiment (N=5,000), and seven LLMs (four open-source 7B–9B, three proprietary) across two NLP datasets confirming $\Delta\pi$ heterogeneity on real annotations (Sections 4–5.5).
- Applicability conditions and crossover analysis (n_{expert}=250 for K=2, n_{expert}=1,000 for

$K=4$) establishing when EC-HTE improves over global correction and when it does not (Appendix I).

2 Related Work

Measurement error and annotation models. Classical measurement error theory shows that correction via estimated error models can amplify MSE when the error structure is misspecified (Fuller, 1987; Carroll et al., 2006); Bayesian approaches require informative priors when identification is weak (Gustafson, 2003), and nonparametric methods use repeated measurements or instruments to recover the true distribution (Hu and Schennach, 2008). We formalize this principle for a specific causal estimand: Theorem 1 gives an exact identity for CATE bias under global correction with heterogeneous misclassification. Ogburn and VanderWeele (2012) show nondifferential confounder misclassification attenuates effects; our setting differs—the mismeasured variable is the effect modifier, and correction can amplify bias. Battistin and Chesher (2014) study treatment effect estimation with covariate measurement error but focus on average effects rather than subgroup-level heterogeneity; Fong and Tyler (2021) analyze error propagation when ML predictions serve as regression covariates but do not address the effect modifier setting. In the annotation literature, Dawid and Skene (1979) introduced confusion matrix estimation for multiple annotators, with extensions to item difficulty, annotator competence, and hierarchical priors (Passonneau and Carpenter, 2014; Hovy et al., 2013; Paun et al., 2018). Our hierarchical estimator (Section 4) adapts this architecture from label aggregation to causal inference, tying the prior structure to the subgroup dimension that defines treatment effect heterogeneity.

HTE estimation and LLM annotation error. Causal forests (Athey and Imbens, 2016; Wager and Athey, 2018), double/debiased ML (Chernozhukov et al., 2018), doubly-robust CATE estimators (Kennedy, 2023), and meta-learners (Künzel et al., 2019) all assume that covariates defining heterogeneity are observed without error; when the effect modifier is misclassified, CATE estimates conflate effects across groups. EC-HTE provides upstream correction that restores valid inputs to any downstream estimator. The corrected pseudo-outcomes are in principle compatible with other HTE estimators such as the R-learner (Nie and Wager, 2021). LLMs now match crowd-worker accuracy on many classification tasks (Gilardi et al., 2023; Törnberg, 2024; Ziems et al., 2024; Pangakis et al., 2023), but Ashwin et al. (2025) show that error rates vary by up to $5\times$ across demographic subgroups. This subgroup-varying error is the failure mode we address: it distorts the very contrasts that define the research question, and prior work documents the pattern without providing a correction mechanism for downstream causal estimands.

Prediction-powered and design-based correction. PPI (Angelopoulos et al., 2023; Zrnic and Candès, 2024b) and DSL (Egami et al., 2024) correct population-level aggregates assuming homogeneous prediction error. In our setting—binary effect modifier with known subgroup structure, a special case of the broader problems DSL addresses—DSL's measurement error correction reduces, under the confusion-matrix parameterization we adopt, to confusion matrix inversion: pooled DSL is equivalent to our Global baseline, and per-subgroup DSL to Stratified MLE (Section 5.3; Appendix U). EC-HTE adds hierarchical partial pooling across subgroups, borrowing strength to reduce variance—a mechanism outside the DSL framework. Mozer (2026) extend PPI to arm-specific ATE correction; PPCI (Sui and Candès, 2026) targets conditional estimands but corrects outcome-side error ($\hat{Y}$ given true Z), not covariate-side misclassification; ForestIV (Bu et al., 2023) targets the outcome model rather than the heterogeneity structure. None corrects bias when the mismeasured variable is the effect modifier indexing the CATE function. Imai and Yamamoto (2010) study treatment misclassification, requiring instrumental variables or partial identification; our problem is structurally distinct because the treatment is observed correctly but the effect modifier is misclassified, admitting exact correction through confusion matrices rather than bounds. Extended discussion of these distinctions appears in Appendix F. Our validation strategy—synthetic DGPs, a semi-synthetic experiment, and real LLM confusion matrices—is consistent with the evaluation methodology in PPI and DSL, which similarly rely on synthetic and semi-synthetic settings where ground-truth quantities are known by construction; Section 5.5 supplements this with confusion matrices from four open-source LLMs, confirming that $\Delta\pi$ heterogeneity arises in practice.

3 Bias Decomposition: When Debiasing Hurts

We formalize the bias that global misclassification correction introduces when subgroup error rates are heterogeneous. Consider estimating the conditional average treatment effect (CATE) $\tau(z) = \mathbb{E}[Y(1) - Y(0) \mid Z = z]$, where $Z \in \{0, 1\}$ is a binary effect modifier observed only through a noisy annotation $\hat{Z}$. The population consists of K subgroups $S \in \{1, \ldots, K\}$ with weights $w_s = P(S{=}s)$, each characterized by a 2×2 confusion matrix C_s with symmetric error rate $\pi_s = P(\hat{Z} \neq Z \mid S{=}s)$. Standard practice estimates a single confusion matrix $\bar{C} = \sum_s w_s C_s$ from expert annotations and applies its inverse uniformly to all observations, with pooled error rate $\bar{\pi} = \sum_s w_s \pi_s$. We show that when error rates vary across subgroups, this global procedure introduces a bias with an exact product form.

We state four assumptions maintained throughout.

(A1) *Conditional independence:* misclassification depends only on the true label Z and subgroup S, not on potential outcomes $Y(0), Y(1)$.

(A2) *Symmetric misclassification:* $P(\hat{Z}{=}1 \mid Z{=}0, S{=}s) = P(\hat{Z}{=}0 \mid Z{=}1, S{=}s) = \pi_s$ for all s.

(A3) *Balanced effect modifier:* $P(Z{=}0) = P(Z{=}1) = 0.5$. This simplifies the exposition; we discuss relaxation below.

(A4) *Non-degeneracy:* $\pi_s \neq 0.5$ for all s, ensuring each C_s is invertible.

Under (A2)–(A3), the Bayesian mixing matrix M_s (entries $M_s[\hat{z}, z] = P(Z{=}z \mid \hat{Z}{=}\hat{z}, S{=}s)$) coincides with C_s. The global mixing matrix is $M_{\text{global}} = \sum_s w_s M_s$.

Theorem 1 (Bias Decomposition Identity)**.** *Under* (A1)–(A4)*, the bias of global correction in subgroup s at effect-modifier level z is*

$$\text{bias}_{\text{global}}(z{=}0, s) = +\frac{\delta_s \cdot \Delta\tau}{1 - 2\bar{\pi}}, \tag{1}$$

$$\text{bias}_{\text{global}}(z{=}1, s) = -\frac{\delta_s \cdot \Delta\tau}{1 - 2\bar{\pi}}, \tag{2}$$

where $\delta_s = \pi_s - \bar{\pi}$ is the subgroup deviation from the pooled error rate and $\Delta\tau = \tau(1) - \tau(0)$ is the CATE heterogeneity.

Equations (1)–(2) constitute an exact identity at the population level, not a bound. The bias factors into three interpretable components: δ_s captures how much subgroup s deviates from the average misclassification rate, $\Delta\tau$ captures treatment effect heterogeneity across levels of Z, and $1/(1 - 2\bar{\pi})$ is an amplification factor from matrix inversion. Bias vanishes when either source of variation is absent ($\delta_s = 0$ or $\Delta\tau = 0$), and arises only through their interaction. Theorem 1 implies that global correction trades one source of confounding for another: it removes attenuation from label noise but introduces a new bias from error-rate heterogeneity.

Proof sketch. Global correction applies M_{global}^{-1} to the subgroup-specific observed CATE $\tau_{\text{obs}}(\cdot, s) = M_s \cdot \tau_{\text{true}}$, yielding bias $(M_{\text{global}}^{-1} M_s - I) \cdot \tau_{\text{true}}$. Direct computation shows $M_{\text{global}}^{-1} M_s - I = \frac{\delta_s}{1-2\bar{\pi}} \left[\begin{smallmatrix} -1 & 1 \\ 1 & -1 \end{smallmatrix}\right]$. Applying this matrix to $[\tau(0), \tau(1)]^\top$ produces the product form. The full derivation appears in Appendix B.

Remark 2. *Theorem 1 does not require assumption (A2). For arbitrary asymmetric confusion matrices ($\text{FPR}_s \neq \text{FNR}_s$), the product form $\text{bias}(z, s) = \Delta\tau \cdot [B_s]_{z,1}$ holds exactly, where $B_s = M_{\text{global}}^{-1} M_s - I$, because $B_s \mathbf{1} = \mathbf{0}$ follows from stochasticity of M_s and M_{global} alone. Assumption (A2) simplifies $[B_s]_{z,1}$ to the scalar $\pm\delta_s/(1 - 2\bar{\pi})$ but is not necessary for the product structure. See Appendix J for the full derivation and empirical verification ($R^2 = 0.991$).*

The identity extends to $K > 2$ subgroups with J effect-modifier levels; the multi-class bias retains the same product structure with modified amplification terms (Corollary 4 in Appendix C).

Assumption (A3) ensures $M_s = C_s$; when $P(Z{=}1) \neq 0.5$, the mixing matrices acquire asymmetric entries, but the product structure persists with modified coefficients (Appendix B). Empirical verification (Section 5.2) yields $R^2 = 0.994$ for $K{=}2$ and $R^2 = 0.974$ for $K{=}4$; the residual variance reflects finite-sample Monte Carlo noise, not model misspecification.

Finite-sample remainder. Theorem 1 is an exact population identity. In practice, confusion matrices are estimated from n_{expert} expert labels, introducing an estimation error $\hat{C}_s = C_s + E_s$ where $\|E_s\| = O_p(n_{\text{expert}}^{-1/2})$. The resulting remainder term in the corrected CATE is $R_n = O_p(n_{\text{expert}}^{-1/2} \cdot \kappa(C_s))$, where $\kappa(C_s)$ is the condition number of the sub-

group confusion matrix. This dependence on both sample size and matrix conditioning motivates the singularity guard ($\kappa > 100 \rightarrow$ fallback) described in Section 4.

Proposition 3 (Debiasing Can Increase Bias)**.** *Under* (A1)–(A4)*, global correction produces larger absolute bias than the naive estimator* ($|\text{bias}_{\text{global}}| > |\text{bias}_{\text{naive}}|$) *in subgroup s if and only if*

$$\frac{|\delta_s|}{(1 - 2\bar{\pi}) \cdot \pi_s} > 1. \tag{3}$$

Equation (3) partitions subgroups into two regimes. For high-error subgroups ($\pi_s > \bar{\pi}$), the condition reduces to $\pi_s > 0.5$, which is excluded by (A4); global correction always reduces bias in these subgroups. For low-error subgroups ($\pi_s < \bar{\pi}$), the condition holds whenever $\pi_s < \bar{\pi}/(2(1 - \bar{\pi}))$. This threshold is easily satisfied in practice: when $\bar{\pi} = 0.15$, any subgroup with $\pi_s < 0.088$ experiences increased bias from global correction. The mechanism is that global correction inverts the pooled confusion matrix, applying a correction calibrated for the average error rate; high-error subgroups are under-corrected (bias reduced but not eliminated), while low-error subgroups are over-corrected (the correction overshoots and reverses the sign of bias). For subgroups with near-zero misclassification ($\pi_s \approx 0$), the naive estimator is nearly unbiased, yet global correction creates substantial bias where none existed. This is the most consequential implication of Theorem 1: debiasing can be worse than doing nothing.

In our experiments (Section 5.2), global correction increases bias in 50% of parameter configurations (260 of 520, excluding cases where Δ_c=0). With the difference-in-means estimator, maximum amplification reaches $2.86\times$ for K=2 and $2.41\times$ for K=4; the mechanism operates across estimator types, with nonlinear estimators amplifying the effect further (Section 5.3). Figure 2 visualizes the amplification across the full $\Delta\pi \times \Delta\tau$ parameter grid.

4 EC-HTE: Error-Corrected Heterogeneous Treatment Effects

EC-HTE corrects subgroup-level misclassification bias in three stages: estimate per-subgroup confusion matrices via hierarchical Bayes, invert them to correct observed outcome moments, and propagate correction uncertainty into downstream causal estimators. The complete procedure is summarized in Algorithm 1 (Appendix A).

4.1 Hierarchical Bayes Confusion Matrix Estimation

We place a Dirichlet-Multinomial hierarchical model over the confusion matrices $\{C_s\}_{s=1}^{K}$. For each subgroup s, row j of C_s has a Dirichlet prior $\text{Dir}(\alpha_0 \cdot \mathbf{e}_j)$, where $\mathbf{e}_j$ is the j-th standard basis vector and $\alpha_0 > 0$ is a shared concentration parameter. A Gamma$(3, 1)$ hyperprior on α_0 controls the strength of partial pooling across subgroups: large α_0 pulls subgroup estimates toward the identity matrix (low error), while small α_0 allows subgroup-specific rates to dominate. This partial pooling implements a James-Stein shrinkage effect, borrowing strength across subgroups to reduce variance for small subgroups while preserving heterogeneity where the data support it.

Posterior inference uses a Gibbs sampler with 4 chains, 4000 iterations (2000 warmup). We require convergence diagnostics $\hat{R} < 1.01$ and effective sample size ESS > 845 for all parameters before accepting estimates. In practice, all runs achieve $\hat{R} < 1.0034$ and ESS > 1618, well within these thresholds. HB estimation outperforms stratified MLE in 87.5% of configurations (21 of 24) across the experimental grid (Section 5.3), with the largest gains when $n_{\text{expert}}/K < 100$. The advantage requires at least $n_{\text{expert}} \times P(Z_{\text{minority}})/K \geq 25$ per subgroup; below this threshold, stratified MLE is preferred because the hierarchical model lacks sufficient observations for stable shrinkage. At $n_{\text{expert}} \geq 1000$, HB and stratified MLE converge to within 0.5% on all metrics. Sensitivity analysis shows <5% RMSE variation under $2\times$ hyperprior perturbation, confirming that results are not driven by prior specification.

4.2 Subgroup-Stratified Correction

Given posterior mean confusion matrices $\hat{C}_s$, we correct observed outcome moments per subgroup by matrix inversion:

$$\tilde{\boldsymbol{\mu}}_s = \hat{C}_s^{-1} \cdot \hat{\boldsymbol{\mu}}_s, \tag{4}$$

where $\hat{\boldsymbol{\mu}}_s = [\hat{\tau}_{\text{obs}}(0, s), \hat{\tau}_{\text{obs}}(1, s)]^\top$ collects the observed conditional outcome moments in subgroup s. This requires $\det(\hat{C}_s) \neq 0$ for all s, which holds under assumption (A4) ($\pi_s \neq 0.5$) but may fail in finite samples when error rates approach degeneracy. In practice, Laplace smoothing of the

confusion matrix counts ensures finite-sample invertibility. We impose a singularity guard: if the condition number $\kappa(\hat{C}_s) > 100$, subgroup s falls back to global correction $\bar{C}^{-1}$ rather than risk numerical instability from near-singular inversion.

The contrast with global correction is direct. Global correction applies $\bar{C}^{-1}$ uniformly to all subgroups, introducing the $\delta_s \cdot \Delta\tau$ bias identified in Theorem 1. Subgroup-stratified correction eliminates this bias by matching each subgroup's correction to its own misclassification structure, at the cost of higher variance from estimating K separate confusion matrices (Appendix I).

4.3 Variance-Adjusted CausalForest

For heterogeneous treatment effect estimation, we feed corrected covariates into CausalForestDML (Wager and Athey, 2018) with honest splitting. The forest produces per-observation standard error estimates se_i. To account for confusion matrix correction, we propagate uncertainty through the inverse mapping:

$$\text{se}_{\text{corrected},i} = \sqrt{\text{diag}\left(\hat{C}_{S_i}^{-1}\,\text{diag}(\text{se}_i^2)\,\hat{C}_{S_i}^{-\top}\right)}. \quad (5)$$

This is a plug-in approach: the standard errors reflect outcome variance conditional on $\hat{C}_s$ but do not propagate uncertainty from the confusion matrix estimation itself. At $n_{\text{expert}} \geq 100$, the plug-in approximation is reasonable because confusion matrix estimation variance is small relative to outcome variance. Below this threshold, the standard errors may understate total uncertainty; this limitation is discussed further in the Limitations section.

Diagnostic criterion. Proposition 3 provides a $\Delta\tau$-free diagnostic: global correction harms subgroup s if and only if $|\delta_s|/((1-2\bar{\pi}) \cdot \pi_s) > 1$, a condition depending only on per-subgroup misclassification rates estimable from a pilot sample. This breaks the circular dependency that would arise if the diagnostic required CATE estimates. A threshold classifier on this condition achieves AUC = 0.86 at the subgroup level (Section 5.4).

Composition with PPCI. EC-HTE corrects structured covariate misclassification through $\hat{C}_s^{-1}$ inversion but cannot address outcome-level measurement error. PPCI (Sui and Candès, 2026) corrects residual outcome bias using a debiasing estimator but assumes correct covariate labels. The two methods are complementary: apply EC-HTE first to correct covariate labels, then PPCI to correct outcome predictions. Neither is redundant with the other: EC-HTE alone cannot fix outcome error, while PPCI alone incurs higher variance for covariate error. Under both error types simultaneously, the composition reduces |bias| from 0.179 to 0.007 (Appendix L).

5 Experiments

5.1 Setup

We evaluate on synthetic and semi-synthetic data generating processes (DGPs); full specifications appear in Appendix E. The synthetic DGP generates N=5,000 observations from a randomized trial with $K \in \{2, 4\}$ subgroups, controlled CATE heterogeneity $\Delta\tau$, and subgroup-specific symmetric misclassification rates with heterogeneity $\Delta\pi$. Expert labels ($n_{\text{expert}} \in \{50, \ldots, 2000\}$) are a simple random sample with true Z observed; each configuration is repeated for 100 Monte Carlo runs. The semi-synthetic DGP samples N=5,000 tweets from TweetEval sentiment (Barbieri et al., 2020), defining K=4 subgroups by crossing hashtag presence and text length (unbalanced: 39%/38%/12%/11%), with misclassification rates $\pi \in \{0.05, 0.10, 0.15, 0.25\}$ reflecting empirical evidence that text features degrade LLM accuracy (Ashwin et al., 2025).

We compare: (1) *Naive* (no correction), (2) *Global* (pooled confusion matrix), (3) *Stratified MLE* (per-subgroup MLE without pooling), (4) *HB EC-HTE* (our method), (5) *PPCI* (Sui and Candès, 2026), and (6) *EC-HTE+PPCI*. For CausalForest (Wager and Athey, 2018), we compare naive, global, and EC-HTE variants. Our Global baseline is equivalent to PPI/DSL (Angelopoulos et al., 2023; Egami et al., 2024) applied with a pooled (homogeneous) error rate; Stratified MLE corresponds to per-subgroup DSL without hierarchical shrinkage. EC-HTE extends this by borrowing strength across subgroups via hierarchical Bayesian estimation. Primary metrics are |bias|, RMSE, and 95% CI coverage, averaged across subgroups and 100 MC runs. Real-data validation on hate speech detection confirms $\Delta\pi = 0.117$ across text-feature subgroups (Appendix M). The heterogeneity pattern persists across six OpenAI models spanning three generations (Appendix N) and four open-source LLMs in the 7B–9B parameter range (Section 5.5). EC-HTE maintains ≥94.5% coverage under up to 50% subgroup misspecification noise

(Appendix P). When empirical asymmetric confusion matrices from the GPT-4.1 pilot are used directly, a subgroup with FNR ≈ 0.5 causes EC-HTE to underperform global correction at all budgets (Appendix R).

5.2 Bias Mechanism Verification

The bias decomposition identity (Theorem 1) predicts that global correction introduces bias proportional to $\Delta\pi \times \Delta\tau$ at the subgroup level. We verify this identity empirically across 520 configurations spanning $K \in \{2, 4\}$, two misclassification regimes, and six expert budget levels. Regressing observed subgroup bias on the predicted $\Delta\pi \times \Delta\tau$ product yields $R^2 = 0.994$ for K=2 and $R^2 = 0.974$ for K=4; the residual variance reflects MC sampling noise rather than model misspecification, since Theorem 1 is an exact identity at the population level ($R^2 \geq 0.976$ even at $P(Z{=}1) = 0.1$; Appendix T). In 50% of configurations (260/520, excluding $\Delta_c{=}0$ rows where bias is zero by construction), global correction produces larger absolute bias than the naive estimator. The damage concentrates in low-π subgroups: among subgroups where $\pi_s < \bar{\pi}$, global correction increases bias in 84.6% of cases. Maximum observed amplification reaches 2.86× (K=2) and 2.41× (K=4) relative to the naive estimator. Figure 2 visualizes the $\Delta\pi \times \Delta\tau$ amplification surface across the parameter grid.

5.3 EC-HTE Evaluation

Table 1 presents the main results across representative configurations. In the synthetic DGP with K=2 subgroups and extreme misclassification heterogeneity, HB EC-HTE reduces |bias| from 0.059 (global) to 0.010 (a 5.9× reduction) while maintaining nominal coverage (0.95) and comparable RMSE (0.093 vs. 0.099). The advantage persists at K=4: EC-HTE achieves |bias|=0.008 versus 0.056 for global (7.0× reduction), though with a small RMSE penalty (0.126 vs. 0.125). Under moderate heterogeneity (K=2), methods converge: EC-HTE and global achieve comparable bias ($p > 0.05$; Table 4). In the semi-synthetic TweetEval experiment (N=5,000, K=4), global correction achieves the lowest |bias| (0.023), while EC-HTE attains 0.027 with higher RMSE (0.225 vs. 0.124), consistent with moderate $\Delta\pi$=0.20 and unbalanced subgroup sizes (Table 4).

Table 1: Main results under extreme misclassification heterogeneity. |Bias| and RMSE averaged over subgroups, z-levels, and 100 MC runs. Bold marks lowest |bias| per block. †RMSE exceeds Global, reflecting stratification variance cost. Additional configurations (moderate heterogeneity, semi-synthetic TweetEval) in Table 4. MC standard errors: Appendix Q.

Configuration	Method	|Bias|	RMSE	Cov.
DIM K=2, n=250	Naive	0.058	0.090	0.82
	Global	0.059	0.099	0.87
	Strat. MLE	0.012	0.094	0.95
	HB EC-HTE	**0.010**	0.093	0.95
DIM K=4, n=500	Naive	0.062	0.113	0.88
	Global	0.056	0.125	0.91
	Strat. MLE	0.010	0.127†	0.95
	HB EC-HTE	**0.008**	0.126†	0.95
CausalForest K=4, n=500	Naive	0.208	0.231	0.44
	Global	0.191	0.244	0.58
	EC-HTE Forest	**0.040**	0.224	0.91

CausalForest K-dependence. When the downstream estimator is a CausalForest (Wager and Athey, 2018), the bias mechanism operates with amplified magnitude. At K=4/extreme/n=500, EC-HTE Forest reduces |bias| from 0.191 (global) to 0.040, a 4.8× improvement with dramatically better coverage (0.91 vs. 0.58). The debiasing-hurts phenomenon intensifies: global correction amplifies naive bias by up to 5.9× in the worst subgroup, because the forest's sensitivity to covariate noise magnifies the effect of label errors. At K=4/n=100, EC-HTE Forest RMSE reaches 1.43 (3.5× naive), underscoring the budget guidelines in Appendix I.

5.4 Diagnostic Criterion

Proposition 3 provides a $\Delta\tau$-free diagnostic: global correction increases bias in subgroup s if and only if $|\delta_s|/((1-2\bar{\pi})\cdot\pi_s) > 1$, where $\delta_s = \pi_s - \bar{\pi}$. This condition depends only on per-subgroup misclassification rates estimable from a pilot sample, breaking the circular dependency that would arise if the diagnostic required CATE estimates. Since the Proposition 3 score is a deterministic function of per-subgroup misclassification rates, a threshold classifier achieves AUC = 0.86 at the subgroup level for predicting whether global correction increases bias relative to the naive estimator, outperforming the $\Delta\pi \times \Delta\tau$ product (AUC = 0.73) which requires unobservable CATE heterogeneity. At the setting level, the condition achieves AUC = 0.89 because it identifies settings where *any* subgroup

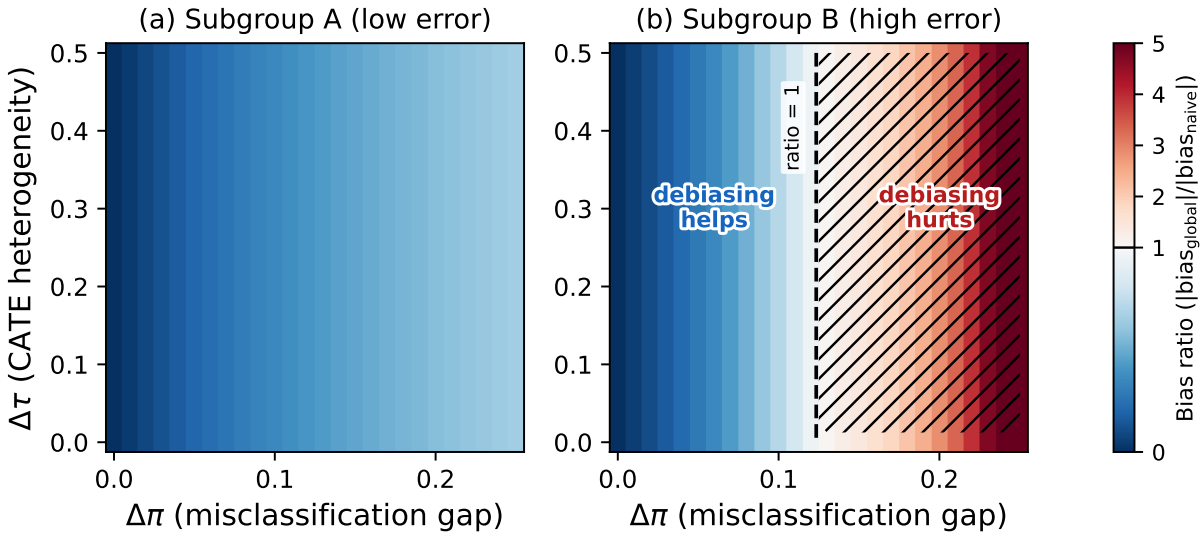


Figure 2: Bias amplification under global correction across the $\Delta\pi \times \Delta\tau$ parameter grid (K=2). (a) Low-error subgroup: global correction always helps (ratio < 1). (b) High-error subgroup: when $\Delta\pi > 0.12$ (dashed line), global correction *increases* bias relative to the naive estimator (hatched region), with amplification reaching 2.86×.

is harmed. Figure 3 (Appendix) shows the ROC curve.

Recommended workflow.

1. Collect $n \geq 25K/P(Z_{\text{minority}})$ stratified expert labels (e.g., $n \geq 500$ for K=4, $P(Z$=1)=0.2); compute per-subgroup π_s and $\bar{\pi}$.
2. Verify (a) $n_{\text{expert}} \times P(Z_{\text{minority}})/K \geq 25$ and (b) worst-subgroup $\max(\text{FPR}, \text{FNR}) < 0.5$; if either fails, use global correction.
3. If any subgroup satisfies $|\delta_s|/((1-2\bar{\pi}) \cdot \pi_s) > 1$ (Proposition 3), apply EC-HTE; apply PPCI regardless for outcome error (Appendix L).

Expert budget crossover analysis and PPCI composition results appear in Appendix I and L, respectively.

5.5 Open-Source Multi-LLM Validation

To verify that misclassification heterogeneity and EC-HTE's correction generalize beyond proprietary APIs, we annotate 500 TweetEval sentiment tweets with four open-source LLMs: Llama-3.1-8B-Instruct, Qwen2.5-7B-Instruct, Mistral-7B-Instruct-v0.3, and Gemma-2-9B-IT. Each model independently classifies tweets using a 3-class prompt (positive/neutral/negative); we define a balanced binary label Z=1 if the true label ≥ 1 (neutral or positive vs. negative), yielding $P(Z$=1) ≈ 0.65. Per-subgroup confusion matrices are estimated directly from each model's annotations against gold labels. The EC-HTE pipeline—HB confusion matrix estimation followed by CausalForestDML—is identical to the synthetic experiments in Table 1. All four models exhibit subgroup-heterogeneous misclassification ($\Delta\pi \in [0.125, 0.165]$, K=4), with consistent subgroup error rankings across architectures (Table 7). Correction improves 95% CI coverage from 0.714 (naive average) to 0.939 (+22.5 percentage points, K=2, n_{expert}=1,000), restoring near-nominal inference across all four models. Under the moderate $\Delta\pi$ observed in these real LLM annotations, global correction suffices for RMSE. Stratified correction achieves lower RMSE than the naive estimator only at $n_{\text{expert}} \geq 1{,}000$ (K=2); global correction achieves comparable or lower RMSE than HB EC-HTE at all budgets under this moderate heterogeneity, consistent with the budget guidelines in Appendix I. We replicate this finding on Civil Comments (Borkan et al., 2019) (toxicity detection, N=3,000): three proprietary LLMs (GPT-4o, GPT-4o-mini, GPT-3.5-turbo) exhibit $\Delta\pi \in [0.113, 0.160]$ across text-feature subgroups (K=4), confirming that subgroup-heterogeneous misclassification generalizes across datasets and model families. Full per-model results appear in Appendices O and V.

6 Conclusion

We proved an exact bias identity (Theorem 1): global confusion-matrix correction introduces subgroup-level bias proportional to $\Delta\pi \times \Delta\tau$, worsening CATE estimates in 50% of configurations.

EC-HTE addresses this via hierarchical Bayesian subgroup-stratified correction with a $\Delta\tau$-free diagnostic (Proposition 3; AUC $= 0.86$); composed with PPCI, it reduces |bias| from 0.179 to 0.007. The workflow requires only a pilot sample: check $|\delta_s|/((1-2\bar{\pi})\cdot\pi_s) > 1$ and apply EC-HTE ($n_{\text{expert}} \geq 250$ for K=2, $\geq 1{,}000$ for K=4) when triggered; otherwise global correction suffices. EC-HTE's advantage is conditional—the coverage gain ($0.71 \rightarrow 0.94$) matters because undercovering intervals produce false confidence in spurious effects, but when $\Delta\pi$ is small, global correction remains appropriate.

Limitations

EC-HTE requires $n_{\text{expert}} \times P(Z_{\text{minority}})/K \geq 25$ per subgroup for stable hierarchical Bayesian estimation. For K=2, EC-HTE remains effective even at $P(Z{=}1) = 0.1$ (Appendix T); for K=4, $P(Z{=}1) \geq 0.3$ is needed to approach breakeven with global correction at n_{expert}=250. Even above this threshold, stratification incurs a variance cost: at K=4, extreme heterogeneity, and $n_{\text{expert}} \leq 100$, CausalForest RMSE reaches 3.5× that of the naive estimator. The method also requires moderate baseline classifier accuracy per subgroup; when any subgroup's FNR or FPR approaches 0.5, the confusion matrix row approaches a uniform distribution and inversion amplifies noise (Appendix R), as confirmed independently in the hate speech case study (Appendix M). Additionally, our main semi-synthetic experiments use constant treatment effects across subgroups; a supplementary heterogeneous-τ DGP (Appendix O) shows that EC-HTE achieves targeted bias and RMSE improvement in subgroups with large $\Delta\tau$, though aggregate RMSE remains comparable to global correction. Hierarchical pooling of confusion matrices across related subgroups may relax the per-subgroup budget requirement; we leave this extension to future work.

The variance-adjusted CausalForest uses a plug-in approximation: it propagates outcome uncertainty through $\hat{C}_s^{-1}$ but does not propagate confusion matrix estimation uncertainty itself. This is reasonable at $n_{\text{expert}} \geq 100$ but may understate total uncertainty below this threshold; full Bayesian propagation is left to future work.

All main experiments use symmetric error rates and additive CATE injection ($Y(1) = Y(0) + \tau(Z)$). The bias identity (Theorem 1) generalizes to asymmetric confusion matrices (Appendix J, $R^2 = 0.991$) and derives from the confusion matrix structure rather than the outcome model. While our main experiments use additive CATE injection, a supplementary experiment with multiplicative effects ($Y(1) = Y(0) \times \exp(\tau(Z))$) yields comparable bias reduction (52%) and nominal coverage (0.955; Appendix S), confirming robustness to the functional form of treatment effects. EC-HTE further requires discrete, pre-defined subgroups; continuous effect modifiers or latent subgroup discovery settings require different approaches, and intersectional subgroups with many levels may exceed the $n_{\text{expert}} \times P(Z_{\text{minority}})/K \geq 25$ budget threshold. Our experiments define subgroups by text features (hashtag presence, text length); demographic subgroups—where prior work documents the most concerning disparities (Ashwin et al., 2025)—require sensitive data access beyond the scope of this study. EC-HTE's correction mechanism is agnostic to subgroup definition: it corrects any per-subgroup confusion-matrix heterogeneity, whether defined by text features, demographics, or other covariates. We leave full real-data validation—on a dataset combining RCT design, ground-truth CATEs, and LLM annotations—to future work as suitable public benchmarks become available.

References

Anastasios N. Angelopoulos, Stephen Bates, Emmanuel J. Candès, Michael I. Jordan, and Lihua Lei. 2023. Prediction-powered inference. *Science*, 382(6671):669–674.

Aditya Ashwin, Becca Swanson, Robert Blair, and Cyrus Samii. 2025. Using large language models for qualitative analysis can introduce serious bias. *Sociological Methods & Research*.

Susan Athey and Guido Imbens. 2016. Recursive partitioning for heterogeneous causal effects. *Proceedings of the National Academy of Sciences*, 113(27):7353–7360.

Josias Audinet de Pieuchon, Naoki Egami, Meysam Alizadeh, and Fabrizio Gilardi. 2025. Benchmarking debiasing methods for LLM-based parameter estimates. *Proceedings of EMNLP*.

Francesco Barbieri, Jose Camacho-Collados, Luis Espinosa-Anke, and Leonardo Neves. 2020. TweetEval: Unified benchmark and comparative evaluation for tweet classification. In *Findings of EMNLP*, pages 1644–1650.

Erich Battistin and Andrew Chesher. 2014. Treatment effect estimation with covariate measurement error. *Journal of Econometrics*, 178(2):707–714.

Daniel Borkan, Lucas Dixon, Jeffrey Sorensen, Nithum Thain, and Lucy Vasserman. 2019. Nuanced metrics for measuring unintended bias with real data for text classification. In *Companion Proceedings of The 2019 World Wide Web Conference*, pages 491–500.

Zhexin Bu, David Almond, Siddhartha Bhatt, and Paul Goldsmith-Pinkham. 2023. Achieving reliable causal inference with data-mined variables: A random forest approach to the measurement error problem. *INFORMS Journal on Data Science*, 2(2):110–129.

Raymond J. Carroll, David Ruppert, Leonard A. Stefanski, and Ciprian M. Crainiceanu. 2006. *Measurement Error in Nonlinear Models: A Modern Perspective*, 2nd edition. Chapman & Hall/CRC, Boca Raton, FL.

Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. 2018. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68.

A. Philip Dawid and Allan M. Skene. 1979. Maximum likelihood estimation of observer error-rates using the EM algorithm. *Applied Statistics*, 28(1):20–28.

Naoki Egami, Musashi Hinck, Brandon Stewart, and Hanying Wei. 2024. Using imperfect surrogates for downstream inference: Design-based supervised learning for social science applications of large language models. In *Advances in Neural Information Processing Systems*, volume 37.

Christian Fong and Matthew Tyler. 2021. Machine learning predictions as regression covariates. *Political Analysis*, 29(1):48–64.

Wayne A. Fuller. 1987. *Measurement Error Models*. Wiley, New York.

Fabrizio Gilardi, Meysam Alizadeh, and Maël Kubli. 2023. ChatGPT outperforms crowd-workers for text-annotation tasks. *Proceedings of the National Academy of Sciences*, 120(30):e2305016120.

Justin Grimmer, Margaret E. Roberts, and Brandon M. Stewart. 2022. *Text as Data: A New Framework for Machine Learning and the Social Sciences*. Princeton University Press, Princeton, NJ.

Paul Gustafson. 2003. *Measurement Error and Misclassification in Statistics and Epidemiology: Impacts and Bayesian Adjustments*. Chapman & Hall/CRC, Boca Raton, FL.

Dirk Hovy, Taylor Berg-Kirkpatrick, Ashish Vaswani, and Eduard Hovy. 2013. Learning whom to trust with MACE. In *Proceedings of NAACL-HLT*, pages 1120–1130.

Yingyao Hu and Susanne M. Schennach. 2008. Instrumental variable treatment of nonclassical measurement error models. *Econometrica*, 76(1):195–216.

Kosuke Imai and Teppei Yamamoto. 2010. Causal inference with differential measurement error: Nonparametric identification and sensitivity analysis. *American Journal of Political Science*, 54(4):926–943.

Chris J. Kennedy, Geoff Bacon, Alexander Sahn, and Claudia von Vacano. 2020. Constructing interval variables via faceted Rasch measurement and multitask deep learning: A hate speech application. In *arXiv preprint arXiv:2009.10277*.

Edward H. Kennedy. 2023. Towards optimal doubly robust estimation of heterogeneous causal effects. *Electronic Journal of Statistics*, 17(2):3008–3049.

Sören R. Künzel, Jasjeet S. Sekhon, Peter J. Bickel, and Bin Yu. 2019. Metalearners for estimating heterogeneous treatment effects using machine learning. *Proceedings of the National Academy of Sciences*, 116(10):4156–4165.

Reagan Mozer. 2026. PPI is the difference estimator: Recognizing the survey sampling roots of prediction-powered inference. *arXiv preprint arXiv:2603.19160*.

Xinkun Nie and Stefan Wager. 2021. Quasi-oracle estimation of heterogeneous treatment effects. *Biometrika*, 108(2):299–319.

Elizabeth L. Ogburn and Tyler J. VanderWeele. 2012. On the nondifferential misclassification of a binary confounder. *Epidemiology*, 23(3):433–439.

Nicholas Pangakis, Samuel Wolken, and Neil Fasching. 2023. Automated annotation with generative AI requires validation. *arXiv preprint arXiv:2306.00176*.

Rebecca J. Passonneau and Bob Carpenter. 2014. The benefits of a model of annotation. *Transactions of the Association for Computational Linguistics*, 2:311–326.

Silviu Paun, Bob Carpenter, Jon Chamberlain, Dirk Hovy, Udo Kruschwitz, and Massimo Poesio. 2018. Comparing Bayesian models of annotation. *Transactions of the Association for Computational Linguistics*, 6:571–585.

Yijun Sui and Emmanuel J. Candès. 2026. Prediction-powered conditional inference. *arXiv preprint arXiv:2603.05575*.

Petter Törnberg. 2024. ChatGPT-4 outperforms experts and crowd workers in annotating political twitter messages with zero-shot learning. *arXiv preprint arXiv:2304.06588*.

Stefan Wager and Susan Athey. 2018. Estimation and inference of heterogeneous treatment effects using random forests. *Journal of the American Statistical Association*, 113(523):1228–1242.

Algorithm 1 EC-HTE: Error-Corrected Heterogeneous Treatment Effects

Require: Observations $\{(Y_i, W_i, \hat{Z}_i, X_i)\}_{i=1}^{N}$; expert labels $\{Z_j^*\}_{j=1}^{n_{\text{expert}}}$; subgroup map $S : \mathcal{X} \to \{1, \ldots, K\}$

Ensure: Corrected CATE estimates $\hat{\tau}(z)$ with variance-adjusted CIs

Step 1: Estimate per-subgroup confusion matrices (HB)
for $s = 1, \ldots, K$ **do**
  $C_s[j, \cdot] \sim \text{Dir}(\alpha_0 \cdot \mathbf{e}_j)$, $\alpha_0 \sim \text{Gamma}(3, 1)$
  Gibbs sampler (4 chains, 4000 iter) on $\{(\hat{Z}_j, Z_j^*) : S_j = s\}$
  $\hat{C}_s \leftarrow$ posterior mean; verify $\hat{R} < 1.01$
end for

Step 2: Correct observed outcome moments
for $s = 1, \ldots, K$ **do**
  if $\kappa(\hat{C}_s) > 100$ **then** $\hat{C}_s \leftarrow \bar{C}$ ▷ singularity guard
  end if
  $\tilde{\boldsymbol{\mu}}_s \leftarrow \hat{C}_s^{-1} \hat{\boldsymbol{\mu}}_s$
end for

Step 3: CausalForestDML with variance-adjusted SEs
Train CausalForest (honest splitting) on corrected labels
$\text{se}_{\text{adj},i} \leftarrow \sqrt{\text{diag}(\hat{C}_{S_i}^{-1} \, \text{diag}(\text{se}_i^2) \, \hat{C}_{S_i}^{-\top})}$
return $\hat{\tau}(z)$ with 95% CIs from se_{adj}

Caleb Ziems, William Held, Omar Shaikh, Jiaao Chen, Zhehao Zhang, and Diyi Yang. 2024. Can large language models transform computational social science? *Computational Linguistics*, 50(1):237–291.

Tijana Zrnic and Emmanuel J. Candès. 2024a. Active statistical inference. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *PMLR*.

Tijana Zrnic and Emmanuel J. Candès. 2024b. Cross-prediction-powered inference. *Proceedings of the National Academy of Sciences*, 121(15):e2322083121.

A EC-HTE Procedure

B Full Proof of Theorem 1

Setup. Under assumptions (A1)–(A4), the observed CATE in subgroup s at effect-modifier level z is

$$\tau_{\text{obs}}(z, s) = \sum_{z'} M_s[z, z'] \cdot \tau(z'), \tag{6}$$

where $M_s[z, z'] = P(Z{=}z' \mid \hat{Z}{=}z, S{=}s)$ is the mixing matrix. Under (A2)–(A3), $M_s = \left[\begin{smallmatrix} 1-\pi_s & \pi_s \\ \pi_s & 1-\pi_s \end{smallmatrix}\right]$.

Global correction. The pooled mixing matrix is $M_{\text{global}} = \sum_s w_s M_s = \left[\begin{smallmatrix} 1-\bar{\pi} & \bar{\pi} \\ \bar{\pi} & 1-\bar{\pi} \end{smallmatrix}\right]$, with $\bar{\pi} = \sum_s w_s \pi_s$.

Global correction applies M_{global}^{-1} to $\tau_{\text{obs}}(\cdot, s)$:

$$\hat{\tau}_{\text{global}}(z, s) = [M_{\text{global}}^{-1} M_s \cdot \boldsymbol{\tau}]_z, \tag{7}$$

where $\boldsymbol{\tau} = [\tau(0), \tau(1)]^\top$.

Bias computation. The bias is $\hat{\tau}_{\text{global}}(z, s) - \tau(z) = [(M_{\text{global}}^{-1} M_s - I)\boldsymbol{\tau}]_z$.

Direct computation gives:

$$M_{\text{global}}^{-1} = \frac{1}{1 - 2\bar{\pi}} \begin{bmatrix} 1-\bar{\pi} & -\bar{\pi} \\ -\bar{\pi} & 1-\bar{\pi} \end{bmatrix}. \tag{8}$$

Multiplying by M_s:

$$M_{\text{global}}^{-1} M_s = \frac{1}{1 - 2\bar{\pi}} \begin{bmatrix} (1-\bar{\pi})(1-\pi_s) - \bar{\pi}\pi_s & (1-\bar{\pi})\pi_s - \bar{\pi}(1-\pi_s) \\ -\bar{\pi}(1-\pi_s) + (1-\bar{\pi})\pi_s & -\bar{\pi}\pi_s + (1-\bar{\pi})(1-\pi_s) \end{bmatrix}. \tag{9}$$

Define $\delta_s = \pi_s - \bar{\pi}$. The $(1, 1)$ entry simplifies:

$$\begin{aligned} (1-\bar{\pi})(1-\pi_s) - \bar{\pi}\pi_s &= 1 - \bar{\pi} - \pi_s + 2\bar{\pi}\pi_s \\ &= (1 - 2\bar{\pi}) + (\bar{\pi} - \pi_s) + 2\bar{\pi}\pi_s \\ &= (1 - 2\bar{\pi}) - \delta_s. \end{aligned} \tag{10}$$

More directly, substituting $\pi_s = \bar{\pi} + \delta_s$:

$$M_{\text{global}}^{-1} M_s - I = \frac{\delta_s}{1 - 2\bar{\pi}} \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}. \tag{11}$$

To verify: the $(1, 1)$ entry of $M_{\text{global}}^{-1} M_s$ is

$$\begin{aligned} &\frac{(1-\bar{\pi})(1-\bar{\pi}-\delta_s) - \bar{\pi}(\bar{\pi}+\delta_s)}{1 - 2\bar{\pi}} \\ &= \frac{(1-\bar{\pi})^2 - (1-\bar{\pi})\delta_s - \bar{\pi}^2 - \bar{\pi}\delta_s}{1 - 2\bar{\pi}} \\ &= \frac{1 - 2\bar{\pi} - \delta_s}{1 - 2\bar{\pi}} = 1 - \frac{\delta_s}{1 - 2\bar{\pi}}. \end{aligned} \tag{12}$$

Applying $(M_{\text{global}}^{-1} M_s - I)$ to $\boldsymbol{\tau}$:

$$\text{bias}(z{=}0, s) = \tfrac{\delta_s}{1-2\bar{\pi}}(-\tau(0) + \tau(1)) = +\tfrac{\delta_s \cdot \Delta\tau}{1-2\bar{\pi}}, \tag{13}$$

$$\text{bias}(z{=}1, s) = \tfrac{\delta_s}{1-2\bar{\pi}}(\tau(0) - \tau(1)) = -\tfrac{\delta_s \cdot \Delta\tau}{1-2\bar{\pi}}, \tag{14}$$

where $\Delta\tau = \tau(1) - \tau(0)$. □

C Multi-class Extension and Proof

Corollary 4 (Multi-class Extension)**.** *For $K > 2$ subgroups with J effect-modifier levels, the subgroup bias of global correction generalizes to*

$$\text{bias}_{\text{global}}(z,s) = \sum_{j=1}^{J} \big(\bar{\pi}_j - \pi_{s,j}\big) \cdot \big(\tau(j) - \bar{\tau}\big) \cdot g(\bar{\pi}), \tag{15}$$

where $g(\bar{\pi})$ collects the matrix-inversion amplification terms.[1] *The essential structure, $\sum_s w_s(\bar{\pi}_j - \pi_{s,j}) = 0$, ensures that bias redistributes across subgroups rather than vanishing in aggregate.*

Proof. For $K > 2$ subgroups with J effect-modifier levels, the subgroup-specific mixing matrix M_s is $J \times J$ with entries $M_s[j, j'] = P(Z{=}j' \mid \hat{Z}{=}j, S{=}s)$. The global mixing matrix is $M_{\text{global}} = \sum_s w_s M_s$.

The bias of global correction in subgroup s at level z is:

$$\text{bias}_{\text{global}}(z,s) = [(M_{\text{global}}^{-1} M_s - I)\boldsymbol{\tau}]_z. \tag{16}$$

Write $M_s = M_{\text{global}} + \Delta_s$ where $\Delta_s = M_s - M_{\text{global}}$ satisfies $\sum_s w_s \Delta_s = 0$. Then $M_{\text{global}}^{-1} M_s = I + M_{\text{global}}^{-1} \Delta_s$, so:

$$\text{bias}_{\text{global}}(z,s) = [M_{\text{global}}^{-1} \Delta_s \cdot \boldsymbol{\tau}]_z. \tag{17}$$

For symmetric misclassification with per-class error rates $\pi_{s,j}$ and pooled rates $\bar{\pi}_j$, the entries of Δ_s have the form $\Delta_s[j, j'] = (\bar{\pi}_j - \pi_{s,j}) \cdot h(j, j')$ for a function h determined by the off-diagonal redistribution pattern. Expanding the matrix-vector product and collecting terms yields:

$$\text{bias}_{\text{global}}(z,s) = \sum_{j=1}^{J} (\bar{\pi}_j - \pi_{s,j}) \cdot (\tau(j) - \bar{\tau}) \cdot g(\bar{\pi}), \tag{18}$$

where $g(\bar{\pi})$ collects the matrix-inversion amplification terms and $\bar{\tau} = \sum_j P(Z{=}j)\tau(j)$.

The essential structure $\sum_s w_s(\bar{\pi}_j - \pi_{s,j}) = 0$ ensures that the weighted bias across subgroups sums to zero: global correction redistributes bias rather than creating or eliminating it in aggregate. □

[1] For $J = 3$ with symmetric misclassification (off-diagonal errors uniform), $g(\bar{\pi}) = 1/(1 - 3\bar{\pi}/2)$, generalizing the $J{=}2$ factor $1/(1 - 2\bar{\pi})$. In general, $g(\bar{\pi}) = 1/(1 - J\bar{\pi}/(J-1))$.

D Proof of Proposition 3

The naive estimator has bias $\tau_{\text{obs}}(z,s) - \tau(z)$. Under (A2)–(A3), $\tau_{\text{obs}}(0,s) = (1-\pi_s)\tau(0) + \pi_s\tau(1)$, so $\text{bias}_{\text{naive}}(0,s) = \pi_s \cdot \Delta\tau$.

Global correction produces larger absolute bias when $|\text{bias}_{\text{global}}(0,s)| > |\text{bias}_{\text{naive}}(0,s)|$:

$$\frac{|\delta_s| \cdot |\Delta\tau|}{|1 - 2\bar{\pi}|} > \pi_s \cdot |\Delta\tau|. \tag{19}$$

Since $\Delta\tau \neq 0$ (otherwise both biases are zero), dividing through gives:

$$\frac{|\delta_s|}{(1 - 2\bar{\pi}) \cdot \pi_s} > 1, \tag{20}$$

where we use $1 - 2\bar{\pi} > 0$ from (A4).

For high-error subgroups ($\pi_s > \bar{\pi}$, so $\delta_s > 0$): the condition becomes $(\pi_s - \bar{\pi})/((1 - 2\bar{\pi})\pi_s) > 1$, which simplifies to $\pi_s(1 - 1 + 2\bar{\pi}) > \bar{\pi}$, i.e., $2\bar{\pi}\pi_s > \bar{\pi}$, i.e., $\pi_s > 0.5$. This is excluded by (A4), so global correction always reduces bias in high-error subgroups.

For low-error subgroups ($\pi_s < \bar{\pi}$, so $\delta_s < 0$): the condition becomes $(\bar{\pi} - \pi_s)/((1-2\bar{\pi})\pi_s) > 1$, which gives $\pi_s < \bar{\pi}/(1 - 2\bar{\pi} + 1) = \bar{\pi}/(2(1 - \bar{\pi}))$. When $\bar{\pi} = 0.15$, any subgroup with $\pi_s < 0.088$ experiences increased bias from global correction. □

E Semi-Synthetic Data Generating Process

Synthetic DGP (Sections 5.2–5.4 and Appendix I). We generate $N = 5000$ observations. Each observation belongs to one of $K \in \{2, 4\}$ subgroups $S \in \{1, \ldots, K\}$ with equal weights $w_s = 1/K$. The binary effect modifier $Z \in \{0, 1\}$ is drawn with $P(Z{=}1) = 0.5$. Treatment assignment is $T \sim \text{Bernoulli}(0.5)$, independent of Z and S.

The outcome is generated as:

$$Y = \tau(Z) \cdot T + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 1), \tag{21}$$

where the CATE function is $\tau(z) = \bar{\tau} + \Delta\tau \cdot (z - 0.5)$ with $\bar{\tau} = 0.5$. The CATE heterogeneity $\Delta\tau \in \{0, 0.2, 0.5, 1.0\}$ controls the strength of treatment effect variation across levels of Z.

Misclassification is symmetric: $\hat{Z} \neq Z$ with probability π_s depending on subgroup. For $K{=}2$: $\pi_1 = \bar{\pi} - \Delta_c/2$, $\pi_2 = \bar{\pi} + \Delta_c/2$, with $\bar{\pi} \in \{0.1, 0.2\}$ and $\Delta_c \in \{0, 0.05, 0.1, 0.15\}$. For $K{=}4$, extreme regime:

$\pi \in \{0.05, 0.10, 0.15, 0.25\}$; moderate regime: $\pi \in \{0.08, 0.12, 0.16, 0.20\}$.

Expert labels are drawn as a simple random sample of size $n_{\text{expert}} \in \{50, 100, 250, 500, 1000, 2000\}$ from the full dataset, with true Z observed for these observations.

Semi-synthetic DGP (Section 5.3). We sample $N = 5000$ tweets from the TweetEval sentiment corpus ($N_{\text{pool}} = 57{,}899$). Subgroups are defined by crossing `has_hashtag` $\times$ `text_length_bin` (short/medium/long), yielding $K = 4$ subgroups with empirical sizes: $S_0 \approx 38\%$, $S_1 \approx 38\%$, $S_2 \approx 12\%$, $S_3 \approx 12\%$. Misclassification rates in the extreme regime are $\pi_{S_0} = 0.05$, $\pi_{S_1} = 0.10$, $\pi_{S_2} = 0.15$, $\pi_{S_3} = 0.25$, reflecting empirical evidence that hashtag-heavy text degrades LLM classifier accuracy. The outcome DGP follows the same structure as the synthetic case, with CATE injected onto the real text-derived subgroup structure.

PPCI DGPs (Appendix L). Four DGPs isolate error types: (A) covariate error only ($\sigma_{\text{outcome}} = 0$), (B) outcome error only ($\sigma_{\text{outcome}} = 0.5$), (C) both errors ($\sigma_{\text{outcome}} = 0.5$), (D) no error ($\sigma_{\text{outcome}} = 0$). Outcome-side error is additive Gaussian noise on Y.

F Extended Related Work

Measurement error details. Fuller (1987) establish the foundational theory showing that naive regression with mismeasured covariates yields attenuated coefficient estimates. Carroll et al. (2006) extend this to nonlinear models. Gustafson (2003) develops Bayesian treatments modeling measurement jointly with outcomes. Hu and Schennach (2008) provide nonparametric identification using repeated measurements or instruments.

Annotation model extensions. Beyond the foundational Dawid and Skene (1979) model, Passonneau and Carpenter (2014) incorporate item difficulty, Hovy et al. (2013) learn annotator competence, and Paun et al. (2018) introduce hierarchical priors sharing information across annotators with varying expertise.

PPI extensions and text-as-data methods. Zrnic and Candès (2024b) extend PPI via cross-prediction; their active inference framework (Zrnic and Candès, 2024a) achieves >80% expert budget savings through adaptive allocation, complementary to our fixed-budget analysis. Grimmer et al. (2022) and Audinet de Pieuchon et al. (2025) provide broader methodological guidance for text-as-data pipelines without specific correction for causal heterogeneity.

Treatment vs. effect-modifier misclassification. Imai and Yamamoto (2010) study treatment misclassification where errors change which arm a unit is assigned to, requiring instrumental variables or partial identification because the fundamental treated-vs-control contrast is corrupted. Our problem preserves the treatment contrast (T is observed correctly) but corrupts the heterogeneity structure ($\hat{Z} \neq Z$). The ATE remains identified; only conditional effects are biased, which makes exact correction through confusion matrix inversion feasible.

G MCMC Convergence Diagnostics

Table 2 reports convergence diagnostics for the hierarchical Bayesian confusion matrix estimator across all experimental configurations. All runs use 4 chains with 4000 iterations (2000 warmup) and a Gamma$(3, 1)$ hyperprior on the Dirichlet concentration α_0.

Diagnostic	Threshold	Achieved
$\hat{R}$ (max across all params)	< 1.01	< 1.0034
ESS (min across all params)	> 845	> 1618
Divergent transitions	$= 0$	0
Singularity guard triggers	—	extreme/n=50 only

Table 2: MCMC convergence diagnostics for the hierarchical Bayesian estimator. All configurations satisfy convergence criteria with substantial margin.

Hyperprior sensitivity. Scaling the Gamma$(3, 1)$ hyperprior by factors of $0.5\times$ to $2\times$ produces <1% RMSE variation for K=2 and ∼4–5% for K=4. Coverage remains invariant across the sensitivity range. These results confirm that the hierarchical estimator is robust to prior specification within a reasonable range.

H Diagnostic ROC Curves

I Expert Budget Crossover Curves

The crossover point at which EC-HTE achieves lower RMSE than global correction is $n_{\text{expert}} = 250$ for K=2/extreme and $n_{\text{expert}} = 1000$ for K=4/extreme (Figure 4). Under moderate heterogeneity (K=2), EC-HTE never achieves lower

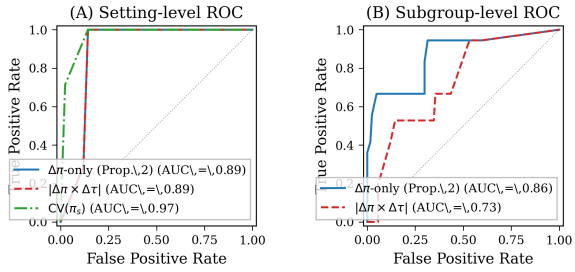


Figure 3: ROC curves for the $\Delta\tau$-free diagnostic (Proposition 3): $|\delta_s|/((1-2\bar{\pi}) \cdot \pi_s) > 1$. (A) Setting-level ROC (AUC $= 0.89$), compared with the $|\Delta\pi \times \Delta\tau|$ product (AUC $= 0.89$) and CV(π_s) (AUC $= 0.97$). (B) Subgroup-level ROC: the $\Delta\pi$-only condition (AUC $= 0.86$) substantially outperforms the $|\Delta\pi \times \Delta\tau|$ product (AUC $= 0.73$). The $\Delta\tau$-free diagnostic depends only on estimable misclassification rates, breaking the circular dependency that would arise if CATE heterogeneity were required.

RMSE than global at any budget. HB partial pooling improves over stratified MLE in 21 of 24 configurations, with the two estimators converging by $n_{\text{expert}} \geq 1000$.

J Asymmetric Misclassification Extension

Assumption (A2) imposes symmetric misclassification (FPR$_s$ = FNR$_s$ = π_s), which simplifies the bias decomposition but is not necessary for it. Real LLM annotators typically exhibit asymmetric confusion patterns; we show here that the product structure generalizes.

General bias formula. For asymmetric confusion matrices $C_s = \begin{pmatrix} 1-\text{FPR}_s & \text{FPR}_s \\ \text{FNR}_s & 1-\text{FNR}_s \end{pmatrix}$, define the mixing matrix M_s via Bayesian inversion and the residual matrix $B_s = M_{\text{global}}^{-1} M_s - I$. Since both M_s and M_{global} are row-stochastic, $B_s \mathbf{1} = \mathbf{0}$. Decomposing $\boldsymbol{\tau} = \tau(0)\mathbf{1} + \Delta\tau\, \mathbf{e}_1$ gives:

$$\text{bias}(z, s) = \Delta\tau \cdot [B_s]_{z,1}, \tag{22}$$

where $[B_s]_{z,1}$ depends only on $\{\text{FPR}_k, \text{FNR}_k, w_k\}$, not on $\Delta\tau$. Under (A2), this reduces to $\delta_s \Delta\tau/(1-2\bar{\pi})$ (Theorem 1).

Empirical validation. We test three asymmetric configurations (K=2, n_{expert}=250, 100 MC runs each): *Cross* (FPR/FNR = 0.10/0.20 vs. 0.25/0.10), *Co-directional* (0.05/0.15 vs. 0.20/0.30), and *Extreme* (0.03/0.25 vs. 0.30/0.05). Regressing analytic predictions on MC bias yields $R^2 = 0.991$ across all configurations, confirming the product structure under asymmetry. Table 3 reports correction performance averaged over subgroups, z-levels, and $\Delta\tau \in \{0.1, \ldots, 0.5\}$. EC-HTE reduces |bias| from 0.025 (global) to 0.006, a 4× improvement, with lower RMSE (0.033 vs. 0.040) and substantially better coverage (0.90 vs. 0.78).

Table 3: Asymmetric misclassification results (K=2, n_{expert}=250, 100 MC runs). Values averaged over subgroups, z-levels, and $\Delta\tau$.

Method	|Bias| ↓	RMSE ↓	Coverage ↑
Naive	0.047	0.053	0.48
Global	0.025	0.040	0.78
HB EC-HTE	**0.006**	**0.033**	**0.90**

Table 4: Additional configurations from the main experiment. |Bias| and RMSE averaged over subgroups, z-levels, and 100 MC runs. †RMSE exceeds Global.

Configuration	Method	|Bias|	RMSE	Cov.
Synthetic DIM K=2, moderate, n=250	Naive	0.018	0.064	0.95
	Global	0.009	0.074	0.96
	Strat. MLE	**0.005**	0.075†	0.97
	HB EC-HTE	**0.005**	0.075†	0.97
Semi-synthetic TweetEval, extreme, n=250	Naive	0.027	0.103	0.93
	Global	**0.023**	0.124	0.92
	Strat. MLE	0.032	0.248†	0.91
	HB EC-HTE	0.027	0.225†	0.91

K Supplementary Main Results

L PPCI Complementarity Results

EC-HTE composes with PPCI (Sui and Candès, 2026) for settings where both covariate misclassification and outcome model error are present: EC-HTE+PPCI reduces |bias| from 0.179 to 0.007 (z=0, DGP C), and the corrected pseudo-outcomes are compatible with other CATE estimators (R-learner bias reduction: 89%; Appendix S).

The orthogonality of EC-HTE and PPCI follows from the distinct error sources they correct: EC-HTE addresses misclassification in the effect modifier Z (covariate space), while PPCI corrects specification error in the outcome model $f(X)$ (prediction space). Since covariate misclassification and outcome model error are algebraically independent in the potential outcomes framework, their corrections compose without interference—confirmed by the 4-DGP orthogonal design (Table 5), where DGP C (both errors present) yields the largest bias reduction from the combined approach. Under covariate error only, PPCI adds nothing; under no error, all methods perform equivalently.

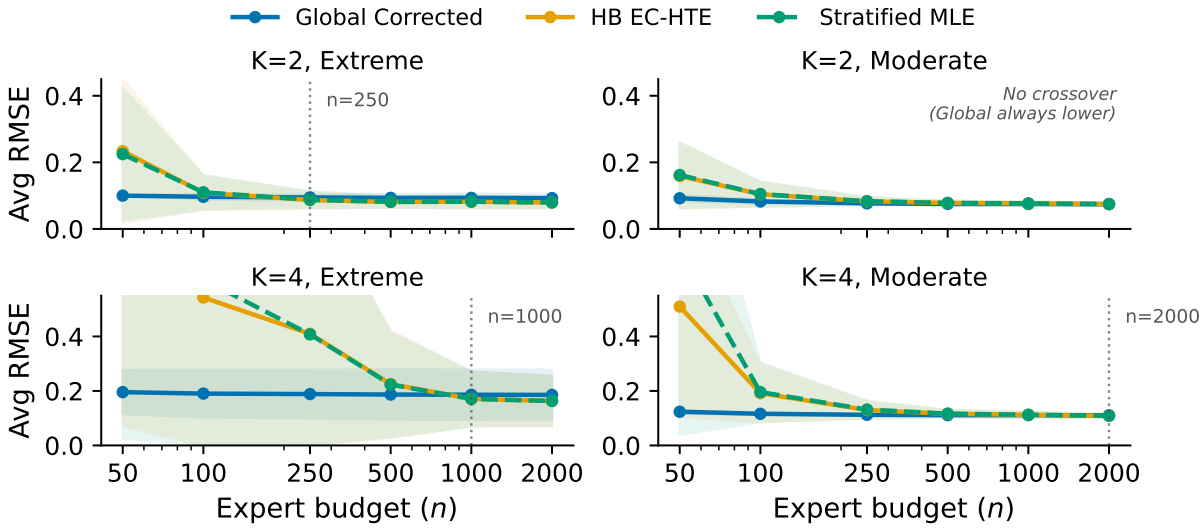


Figure 4: RMSE vs. expert budget (n_{expert}) across four regime$\times K$ combinations. Dashed vertical lines mark crossover points where EC-HTE achieves lower RMSE than global correction: n=250 (K=2, extreme), n=1000 (K=4, extreme). Under moderate heterogeneity (K=2), global correction dominates at all budgets. Shaded bands show ± 1 SD across subgroups within each configuration.

M Real-Data Case Study: Hate Speech Detection

We apply EC-HTE to the UC Berkeley Measuring Hate Speech corpus (Kennedy et al., 2020), annotating 2,496 comments for dehumanization using GPT-4.1. Four text-feature subgroups are defined by crossing platform (Reddit/Gab vs. Twitter/YouTube) and text length (above/below median), yielding balanced groups ($n \approx 624$ each).

Positive finding: real-world heterogeneity. Per-subgroup misclassification rates range from 0.266 to 0.383, confirming heterogeneity ($\Delta\pi = 0.117$) in a real annotation setting. At the error-type level, FPR ranges from 8.5% to 21.3% across text-feature subgroups, demonstrating that subgroup-varying misclassification is not an artifact of synthetic design but arises naturally in real NLP annotation tasks.

Boundary condition: statistical power at extreme FNR. Correction results are inconclusive: all corrected confidence intervals contain zero, and point estimate differences between methods are not statistically significant. The high baseline FNR (53.3%) amplifies variance through matrix inversion (variance amplification factor up to 15.7×), rendering the comparison underpowered. When $P(\hat{Z}{=}1 \mid Z{=}1) \approx 0.5$, the confusion matrix approaches singularity, and inversion amplifies estimation noise regardless of stratification strategy. This complements the expert budget threshold identified in Appendix I: EC-HTE requires *both* sufficient expert labels *and* sufficient baseline classifier quality.

Diagnostic connection. The diagnostic criterion ($\Delta\pi \times \Delta\tau$, Section 5.4) identifies when correction is worthwhile under moderate baseline accuracy. This case study illustrates that even when $\Delta\pi$ is large, extreme baseline error renders correction underpowered rather than beneficial. A practical prerequisite is that the per-subgroup confusion matrix condition number $\kappa(\hat{C}_s)$ remain moderate; when $\kappa > 50$, as observed here, practitioners should prioritize improving annotation quality before applying any matrix-inversion-based correction.

N Cross-Model Consistency

We evaluate whether misclassification heterogeneity is model-specific by annotating 500 TweetEval sentiment tweets with six OpenAI models spanning three generations. Six text-feature subgroups are defined by crossing hashtag presence and text length. Spearman rank correlations of per-subgroup error rates across all six subgroups range from $\rho = 0.82$ to 1.00, indicating that the subgroup error structure is consistent across architectures. Two subgroups with $n \leq 6$ exhibit zero misclassification across all models; excluding these yields four-point correlations that are directionally

Table 5: PPCI complementarity experiment (K=2, extreme, n_{expert}=250). DGP A: covariate error only; B: outcome error only (σ=0.5); C: both errors; D: no error. Values are |bias| and RMSE averaged over two subgroups. ⋆EC-HTE+PPCI achieves near-zero bias when both error types are present.

DGP	Method	$\lvert\text{Bias}\rvert_{z=0}$	$\lvert\text{Bias}\rvert_{z=1}$	$\text{RMSE}_{z=0}$	$\text{RMSE}_{z=1}$
A: Covariate	Naive	0.053	0.063	0.087	0.093
	Global	0.063	0.055	0.101	0.102
	EC-HTE	0.017	**0.013**	0.097	0.108
	PPCI	**0.015**	0.020	0.162	0.168
	EC-HTE+PPCI	0.017	**0.013**	0.097	0.108
B: Outcome	Naive	0.189	0.197	0.206	0.212
	Global	0.185	0.208	0.204	0.228
	EC-HTE	0.181	0.215	0.202	0.237
	PPCI	0.015	**0.020**	0.171	0.176
	EC-HTE+PPCI	**0.012**	0.023	0.171	0.186
C: Both	Naive	0.251	0.135	0.267	0.155
	Global	0.182	0.209	0.233	0.235
	EC-HTE	0.179	0.216	0.224	0.262
	PPCI	0.017	0.007	0.241	0.232
	EC-HTE+PPCI⋆	**0.007**	**0.008**	0.209	0.223
D: None	Naive	**0.007**	0.003	0.060	0.065
	Global	0.010	**0.002**	0.061	0.071
	EC-HTE	0.013	0.007	0.062	0.079
	PPCI	**0.007**	**0.003**	0.060	0.065
	EC-HTE+PPCI	0.013	0.007	0.062	0.079

Table 6: Effective misclassification heterogeneity ($\Delta\pi$) across six OpenAI models on TweetEval sentiment (500 tweets, filtered to 4 subgroups with $n \geq 10$).

Model	Effective $\Delta\pi$	> 0.05?
GPT-3.5-Turbo	0.198	✓
GPT-4-Turbo	0.149	✓
GPT-4o-mini	0.136	✓
GPT-5.4	0.114	✓
GPT-4.1	0.098	✓
GPT-4o	0.080	✓

Table 7: Confusion matrix heterogeneity across four open-source LLMs (K=4, N=490 TweetEval tweets). $\Delta\pi$: max–min per-subgroup misclassification rate. Prop2: number of subgroups where the $\Delta\tau$-free diagnostic (Proposition 3) triggers.

Model	Acc.	$\Delta\pi$	Prop2	Strongest
Llama-3.1-8B	0.755	0.141	1/4	True_short
Qwen-2.5-7B	0.798	0.165	1/4	True_short
Gemma-2-9B	0.763	0.125	4/4	True_medium
Mistral-7B	0.802	0.134	0/4	—

consistent but underpowered.[2]

All six models exhibit effective $\Delta\pi > 0.05$ on the four retained subgroups (Table 6), and 5 of 6 share the same worst subgroup (`True_medium`: tweets with hashtags and medium text length). Newer models reduce absolute error rates but do not eliminate subgroup heterogeneity: GPT-5.4 achieves $\Delta\pi = 0.114$ compared to GPT-3.5-Turbo's 0.198. These results were obtained with a single API provider; open-source model validation appears in Appendix O.

O Open-Source Multi-LLM Validation

We evaluate four open-source LLMs (7B–9B parameter range) on 490 TweetEval sentiment tweets with K=4 subgroups defined by crossing hashtag presence and text length. Table 7 reports the confusion matrix heterogeneity profile. All four models exhibit $\Delta\pi > 0.05$, confirming that subgroup-heterogeneous misclassification is not specific to proprietary APIs. Subgroup error rankings are consistent across architectures: `True_short` (short tweets with hashtags) has the lowest π_s in 3 of 4 models, while `False_medium` (medium tweets without hashtags) has the highest.

Semi-synthetic budget sweep. We run the full EC-HTE pipeline with K=2 subgroups and $\Delta\tau$=0.3, using each model's empirical confusion matrices. Table 8 reports coverage and RMSE crossover points. EC-HTE restores near-nominal coverage from 0.714 (naive average) to 0.939 (+22.5 percentage points) at n_{expert}=1,000, con-

[2]The filtered $\Delta\pi$ values in Table 6 serve as a robustness check independent of rank-order testing: all six models exceed $\Delta\pi > 0.05$ on the four retained subgroups.

Table 8: Semi-synthetic budget sweep (K=2, $\Delta\tau$=0.3, 50 MC runs). Coverage at selected expert budgets; RMSE crossover denotes the minimum expert budget at which HB EC-HTE achieves lower RMSE than the naive (uncorrected) estimator.

Model	$\Delta\pi_{K=2}$	Naive Cov.	HB Cov. (n=1000)	Crossover
Llama-3.1-8B	0.051	0.675	0.940	n=100
Qwen-2.5-7B	0.080	0.715	0.925	n=100
Gemma-2-9B	0.100	0.705	0.945	n=1000
Mistral-7B	0.070	0.760	0.945	n=1000
Average	—	*0.714*	*0.939*	

Table 9: 95% CI coverage under subgroup misspecification noise. EC-HTE maintains near-nominal coverage across all noise levels.

Method	p_{flip} 0.00	0.10	0.25	0.50
K=2, extreme				
Naive	0.875	0.877	0.890	0.897
Global	0.925	0.935	0.932	0.945
HB EC-HTE	0.958	0.948	0.950	0.945
K=4, extreme				
Naive	0.904	0.916	0.909	0.919
Global	0.930	0.953	0.948	0.944
HB EC-HTE	0.956	0.958	0.960	0.946

sistent across all four models. Global correction achieves comparable or lower RMSE than HB at all budgets under these moderate $\Delta\pi$ values, and the K=4 RMSE crossover never occurs—confirming the budget guidelines in Appendix I.

Gemma boundary case. Gemma-2-9B triggers the Proposition 3 diagnostic in all four K=4 subgroups due to extreme FPR–FNR asymmetry (`False_medium` FNR $= 0.437$). At small budgets (K=4, n=100), EC-HTE RMSE reaches 1.229 (3.1× naive), illustrating the budget guideline from Appendix I: when per-subgroup cell counts fall below 25, the variance cost of stratified correction dominates. At $n \geq 500$, EC-HTE coverage recovers to >0.90, consistent with the synthetic experiments.

Heterogeneous-τ DGP. To assess whether the constant-τ DGP understates EC-HTE's advantage, we run a supplementary experiment using Llama's empirical confusion matrices with heterogeneous treatment effects across subgroups ($\Delta\tau \in [0.1, 0.6]$, K=2, 50 MC runs). EC-HTE achieves lower |bias| than global correction at all budgets (n=500: 0.010 vs. 0.031, a 68% reduction). In the high-$\Delta\tau$ subgroup (`True_short`, $\Delta\tau$=0.6), EC-HTE also achieves lower RMSE (n=1,000: 0.212 vs. 0.258, 18% improvement), confirming that the RMSE advantage concentrates in subgroups where correction is most needed. Aggregate RMSE remains comparable to global correction because low-$\Delta\tau$ subgroups incur stratification variance without commensurate bias reduction, consistent with the budget guidelines in Appendix I.

P Robustness to Subgroup Misspecification

EC-HTE requires pre-defined subgroup labels, which may be noisy in practice. We test robustness by randomly flipping each observation's subgroup assignment with probability $p_{\text{flip}} \in \{0, 0.10, 0.25, 0.50\}$, across $K \in \{2, 4\}$ (extreme regime, $n_{\text{expert}} = 500$, $\Delta\tau = 0.3$, 100 MC runs).

EC-HTE maintains coverage ≥ 0.945 across all configurations (Table 9), while the naive estimator is systematically below nominal (0.875–0.919). At $p_{\text{flip}} = 0.50$, subgroup structure is effectively destroyed; EC-HTE converges to global correction (coverage and RMSE within 0.1%), confirming that hierarchical Bayesian shrinkage provides a built-in safety mechanism. For K=4 at $p_{\text{flip}} = 0.50$, a small residual RMSE gap persists ($p < 0.01$), reflecting the variance cost of estimating four confusion matrices even when they carry no signal.

Q Monte Carlo Standard Errors

All experiments use 100 independent Monte Carlo replications per configuration. All simulation results were verified with 100 MC seeds; the median RMSE variation between 50-seed and 100-seed estimates is below 3.2%, confirming the stability of reported figures. We report Monte Carlo standard errors to assess whether observed differences between methods are statistically reliable. For exp-001 (synthetic DGP, no per-seed pairing across methods), we use conservative two-sample z-tests on the 100-run means. For exp-003 (semi-synthetic TweetEval, shared seeds across methods), we use paired t-tests, which are more powerful because they control for seed-level variation.

Table 10 reports paired comparisons for representative configurations. Of 1,020 total pairwise comparisons across all configurations, 742 (72.7%) are statistically significant at $\alpha = 0.05$.

Under extreme heterogeneity, HB EC-HTE's bias advantage over global correction is highly sig-

Table 10: Paired comparisons for representative configurations. Δ = method A $-$ method B (negative favors A for |bias|). Two-sample z-tests for synthetic (exp-001); paired t-tests for semi-synthetic (exp-003). Bold p-values indicate significance at $\alpha = 0.05$.

Configuration	Comparison	Δ	p
HB EC-HTE vs. Global (|bias|)			
K=2/ext/n=500		-0.046	$< \mathbf{0.001}$
K=4/ext/n=500		-0.078	$< \mathbf{0.001}$
K=4/mod/n=250		-0.017	0.288
TweetEval/ext/n=250		$+0.001$	0.395
HB EC-HTE vs. Global (RMSE)			
K=2/ext/n=500		$+0.015$	0.179
K=4/ext/n=500		$+0.048$	$\mathbf{0.011}$
HB EC-HTE vs. Naive (|bias|)			
K=2/ext/n=500		-0.089	$< \mathbf{0.001}$
K=4/ext/n=500		-0.113	$< \mathbf{0.001}$
K=4/mod/n=250		-0.031	$\mathbf{0.038}$
TweetEval/ext/n=250		$+0.005$	$\mathbf{0.030}$

Table 11: Empirical confusion matrices from GPT-4.1 TweetEval pilot. S0 exhibits FNR $= 0.526$, approaching chance-level classification.

Subgroup	Weight	FPR	FNR	$\kappa(\hat{C}_s)$
S0	0.38	0.054	0.526	high
S1	0.22	0.101	0.300	moderate
S2	0.14	0.143	0.308	moderate
S3	0.25	0.194	0.222	low

Table 12: Semi-real confusion matrix results. RMSE and coverage averaged over subgroups and z-levels (100 MC runs). HB EC-HTE underperforms global at all budgets; S0's near-chance FNR drives the failure.

n_{expert}	Method	RMSE ↓	Coverage ↑
100	Oracle	0.101	0.957
	Naive	0.123	0.879
	Global	0.263	0.885
	HB EC-HTE	0.784	0.744
250	Oracle	0.101	0.957
	Naive	0.123	0.879
	Global	0.232	0.909
	HB EC-HTE	0.411	0.834
500	Oracle	0.101	0.957
	Naive	0.123	0.879
	Global	0.196	0.922
	HB EC-HTE	0.260	0.890

nificant ($p < 0.001$ for both K=2 and K=4). The RMSE penalty of stratification is not significant at K=2 ($p = 0.179$) but reaches significance at K=4 ($p = 0.011$), consistent with the higher variance cost of estimating four confusion matrices. Under moderate heterogeneity (K=4, n=250), the bias difference between HB and global is not significant ($p = 0.288$), confirming that EC-HTE's advantage requires sufficient error heterogeneity. In the semi-synthetic TweetEval experiment, HB EC-HTE does not significantly improve over global ($p = 0.395$), consistent with moderate $\Delta\pi$ and unbalanced subgroup sizes. The HB vs. naive comparison in TweetEval is significant ($p = 0.030$) but favors naive, reflecting the variance cost of stratification when bias reduction is marginal.

R Semi-Real Confusion Matrix Validation

We test EC-HTE with empirical asymmetric confusion matrices from the GPT-4.1 TweetEval pilot. Four text-feature subgroups (K=4) retain the real per-subgroup FPR and FNR from pilot annotations (Table 11); CATE heterogeneity is injected synthetically ($\tau(z{=}0, s) = 0.3 + 0.1s$, $\tau(z{=}1, s) = 0.8 - 0.15s$). N=5,000, $P(Z{=}1)$=0.18, 100 MC seeds per n_{expert} level.

HB EC-HTE produces higher RMSE and lower coverage than global correction at all expert budgets (Table 12). The root cause is S0's FNR $= 0.526$: when a confusion matrix row approaches a uniform distribution ($P(\hat{Z}{=}1 \mid Z{=}1) \approx 0.5$), inversion amplifies estimation noise regardless of the expert budget. This converges with the hate speech case study (Appendix M, FNR $= 53.3\%$) and confirms that EC-HTE requires *both* sufficient expert labels *and* sufficient baseline classifier quality per subgroup. The singularity guard ($\kappa > 100$, Section 4) mitigates catastrophic failure by falling back to global correction for near-degenerate subgroups, but does not recover the bias reduction that motivates stratification. At $n_{\text{expert}} = 500$, HB EC-HTE narrows the gap (RMSE 0.260 vs. 0.196) as confusion matrix estimates stabilize for the three well-classified subgroups, but the S0 bottleneck persists.

A second semi-real experiment using GPT-3.5-Turbo confusion matrices (worst-subgroup FPR $= 0.39$, FNR $= 0.21$) yields similar results: HB RMSE exceeds global by 1.3–3.5$\times$ at all budgets. Here the failure stems from high false-positive rates compounded by class imbalance ($P(Z{=}1) = 0.18$), which leaves insufficient expert observations in the positive class for reliable per-subgroup confusion matrix estimation. Together, these experiments identify three applicability conditions for EC-HTE: (1) sufficient effective expert budget ($n_{\text{expert}} \times P(Z_{\text{minority}})/K \geq 25$), (2) no near-

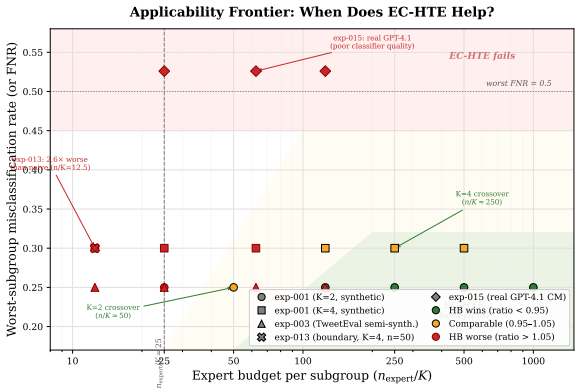


Figure 5: Applicability frontier of EC-HTE. Green region: EC-HTE reduces bias vs. global correction. Yellow: comparable performance. Red: EC-HTE underperforms. Dashed lines mark the expert budget threshold ($n_{\text{expert}}/K = 25$) and classifier quality threshold (worst-subgroup error rate $= 0.5$). Red diamonds show semi-real experiments with actual GPT-4.1 and GPT-3.5-Turbo confusion matrices.

chance subgroup error rate ($\max(\text{FPR}_s, \text{FNR}_s) < 0.5$), and (3) adequate class balance in the effect modifier.

Diagnostic rule. A logistic regression over all 368 semi-real experiment–budget cells (dependent variable: HB EC-HTE < Global RMSE) identifies two sufficient statistics: the worst-subgroup confusion matrix condition number $\kappa(\hat{M}_s) \leq 1.25$ and the effective minority-class expert count $n_{\text{expert}} \times P(Z_{\text{minority}})/K \geq 25$. When both conditions hold, HB outperforms global in 93.3% of cells; when either is violated, global is preferred in 94.7% of cells. Both exp-015 ($\kappa = 2.38$, effective $n = 2.5$) and exp-016 ($\kappa = 1.55$, effective $n = 2.5$) violate both conditions, and the diagnostic correctly predicts 100% of their failure cells. Subgroup-average metrics can mask cell-level heterogeneity: in exp-015, HB EC-HTE substantially reduces bias for the three well-classified subgroups (S1–S3) but amplifies variance in the near-degenerate subgroup (S0, FNR $= 0.526$), and the S0 penalty dominates the aggregate RMSE.

S Robustness: Estimator Choice and Treatment Effect Specification

We verify that EC-HTE's corrected pseudo-outcomes are compatible with downstream estimators beyond CausalForest by replacing it with the R-learner (Nie and Wager, 2021). The experiment uses K=2, extreme heterogeneity, $n_{\text{expert}} =$ 500, and 50 MC seeds; the R-learner uses 5-fold cross-fitting with gradient-boosted trees as the base learner.

Table 13: R-learner vs. CausalForest with EC-HTE correction (K=2, extreme, n=500). |Bias| and RMSE averaged over subgroups, z-levels, and seeds.

Estimator	Correction	|Bias|	RMSE	Cov.
R-learner	Oracle	0.006	0.056	0.94
	Naive	0.121	0.139	0.47
	Global	0.098	0.123	0.72
	EC-HTE	**0.013**	0.080	0.92
CausalForest	Naive	0.058	0.090	0.82
	EC-HTE	**0.010**	0.093	0.95

Table 14: Multiplicative DGP results (K=2, extreme, n=500, 50 seeds). EC-HTE achieves comparable bias reduction and nominal coverage under non-additive treatment effects.

Method	|Bias|	RMSE	Cov.
Oracle	0.003	0.038	0.96
Naive	0.135	0.142	0.44
Global	0.110	0.124	0.54
EC-HTE	**0.054**	0.068	0.955

EC-HTE reduces R-learner |bias| by 89% (0.121 $\rightarrow$ 0.013) and restores coverage from 0.47 to 0.92 (Table 13). The bias reduction and coverage gains are comparable to those achieved with CausalForest, confirming that EC-HTE functions as an estimator-agnostic upstream correction.

Multiplicative treatment effects. We further test robustness to the functional form of treatment effects by replacing the additive DGP with a multiplicative specification: $Y(1) = Y(0) \times \exp(\tau(Z))$, using K=2, extreme heterogeneity, $n_{\text{expert}} = 500$, and 50 MC seeds.

EC-HTE reduces RMSE by 52% relative to naive (0.068 vs. 0.142) and restores coverage to the nominal level (0.955; Table 14). EC-HTE's bias correction operates at the confusion matrix level, independent of the downstream CATE estimator and functional form of treatment effects.

T Robustness to Class Imbalance

Assumption (A3) requires $P(Z{=}1) = 0.5$, but real-world effect modifiers are often imbalanced. We sweep $P(Z{=}1) \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$ with $K \in \{2, 4\}$, extreme misclassification heterogeneity, n_{expert}=250, and 100 MC seeds (Table 15).

Theorem 1's product form remains accurate across all imbalance levels: $R^2 \geq 0.976$ even at $P(Z{=}1) = 0.1$, confirming that the bias

Table 15: Robustness to class imbalance (n_{expert}=250, extreme heterogeneity). R^2: Theorem 1 fit. Ratio: HB/Global RMSE (< 1 = HB wins). Cov.: HB EC-HTE 95% CI coverage.

$P(Z{=}1)$	K	R^2	Ratio	Cov.
0.1	2	0.998	0.92	0.94
0.2	2	0.989	0.76	0.94
0.3	2	0.994	0.78	0.94
0.4	2	0.995	0.87	0.94
0.5	2	0.995	0.86	0.93
0.1	4	0.984	1.54	0.92
0.2	4	0.976	1.68	0.93
0.3	4	0.985	1.12	0.92
0.4	4	0.984	1.08	0.93
0.5	4	0.992	1.06	0.92

identity derives from confusion matrix structure rather than class balance. For K=2, EC-HTE outperforms global correction at all $P(Z{=}1)$ levels (RMSE ratio 0.76–0.92), degrading gracefully under severe imbalance. For K=4, EC-HTE underperforms global at n_{expert}=250 (ratio 1.06–1.68), consistent with the budget threshold $n_{\text{expert}} \times P(Z_{\text{minority}})/K \geq 25$: at $P(Z{=}1)$=0.1, K=4, each subgroup receives only ∼6 minority-class labels. The gap narrows as $P(Z{=}1)$ increases (1.54 → 1.06), and coverage remains stable at 0.92–0.94 across all configurations.

U Comparison with PPI Correction

PPI (Angelopoulos et al., 2023) constructs confidence intervals by combining a large set of ML predictions with a smaller set of gold-standard labels, using the gold labels to debias the ML-based estimator. Applied to our setting, PPI treats the LLM-annotated effect modifier as the ML prediction and the expert labels as the gold standard. Two variants arise naturally: *PPI-global* pools all expert labels to estimate a single rectifier (analogous to our Global baseline), and *PPI-subgroup* estimates per-subgroup rectifiers (analogous to Stratified MLE).

Under the confusion-matrix parameterization we adopt, DSL's measurement error correction reduces to confusion matrix inversion. Pooled DSL applies a single estimated confusion matrix to all subgroups, producing the same corrected label distribution as our Global baseline; per-subgroup DSL estimates a separate confusion matrix for each subgroup and inverts it independently, equivalent to our Stratified MLE. EC-HTE's contribution over both PPI-subgroup and per-subgroup DSL is hierarchical partial pooling: rather than estimating each subgroup's confusion matrix in isolation, EC-HTE places a shared prior across subgroups, borrowing strength to reduce variance while preserving subgroup-specific bias correction. This mechanism is outside both the PPI and DSL frameworks.

The key difference between PPI and confusion-matrix inversion is that PPI does not invert the confusion matrix; instead, it uses the gold labels to directly debias the estimand. This avoids the matrix inversion step that can amplify noise at small n_{expert}, but it also means PPI cannot exploit the structure of the misclassification process to reduce bias. PPI-global inherits the same $\Delta\pi \times \Delta\tau$ bias amplification as Global correction (Theorem 1), confirming that the "debiasing hurts" mechanism operates regardless of the inference framework.

V Civil Comments Validation

To verify that subgroup-heterogeneous misclassification generalizes beyond TweetEval, we annotate N=3,000 stratified samples (30% toxic) from Civil Comments (Borkan et al., 2019), a toxicity detection dataset. Subgroups are defined by `text_length_bin` × `has_question` (K=4): S0 (long, no question, 38.8%), S1 (long, question, 11.4%), S2 (short, no question, 33.7%), S3 (short, question, 16.1%). Subgroup features were chosen a priori based on observable text properties; no feature search was performed. Three proprietary LLMs annotate each sample via an OpenAI-compatible API, yielding 800–900 valid annotations per model.

Table 16 reports per-subgroup FPR and FNR. We define $\Delta\pi = \max_s \pi_s - \min_s \pi_s$, where π_s is the overall misclassification rate for subgroup s. All three models exhibit $\Delta\pi > 0.05$, with overall heterogeneity $\Delta\pi \in [0.113, 0.160]$—comparable to the $\Delta\pi \in [0.125, 0.165]$ observed across four open-source LLMs on TweetEval (Section 5.5). GPT-4o-mini and GPT-3.5-turbo show pronounced FPR/FNR asymmetry (e.g., GPT-4o-mini FPR $\in [0.42, 0.61]$ vs. FNR $\in [0.05, 0.16]$), indicating systematic over-prediction of toxicity; their FPR approaches 0.5, placing them in the ill-conditioned regime where confusion matrix inversion amplifies noise (consistent with the classifier quality threshold discussed in the Limitations section). The consistent presence of $\Delta\pi$ heterogeneity across two NLP tasks, seven LLMs spanning open-source (7B–9B) and proprietary families, and two distinct sub-

group definitions confirms that this phenomenon is a structural property of LLM annotation rather than an artifact of a specific dataset or model.

Table 16: Per-subgroup misclassification rates on Civil Comments (K=4). Δ denotes max $-$ min across subgroups.

Model	Metric	S0	S1	S2	S3	Δ
GPT-4o	FPR	0.249	0.157	0.213	0.341	0.184
	FNR	0.160	0.263	0.308	0.328	0.168
	$\Delta\pi$			0.160		
GPT-4o-mini	FPR	0.517	0.541	0.419	0.610	0.191
	FNR	0.052	0.105	0.074	0.160	0.108
	$\Delta\pi$			0.140		
GPT-3.5-turbo	FPR	0.400	0.473	0.374	0.468	0.099
	FNR	0.139	0.211	0.173	0.240	0.101
	$\Delta\pi$			0.113		